

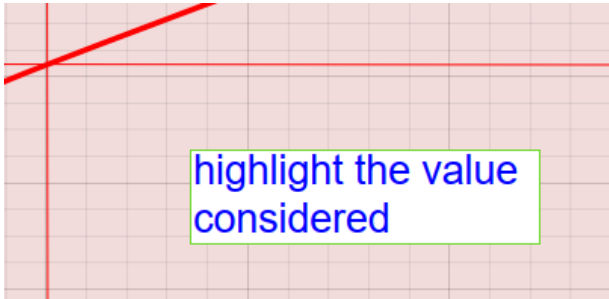
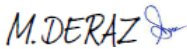
	EPC FOR UPGRADATION OF FACILITIES AT SE FIELD			
	COMMENTS RESOLUTION SHEET (CRS)			
Contract No.	4700022908	Project No.	P30377	Comment Code - 2
Document No.		Transmittal No.	Document Title	☐ 1 Approved – No Comments ☒ 2 Approved Subject to Incorporation of Comments ☐ 3 Revise and Resubmit ☐ 4 Rejected ☐ 5 Retained for Information
P14429-16-99-67-1601	Rev-A	ATC-4700022908-05339-1	TELECOM UPS SIZING CALCULATION - SAHIL AREA	

S. No.	Reference / Page No.	COMPANY Comments	CONTRACTOR Response	Status / Remarks
1.	Page 1 OF 18	Ratings are not approved. Refer comments, rating will change. This shall be subject to COMPANY approval	Noted and updated based on the verified and confirmed ratings dated 19 th of Sep. 2025	
2	Page 8 OF 18	Add Datasheet UPS	Noted and added.	
3.	Page 14 OF 18	Rating shall not be less than the FEED / SOW or Contractual documents	Noted and ratings updated based on the verified and confirmed ratings dated 19 th of Sep. 2025	
4.	Appendix-1.1	Load list shall be based on the final vendor data RSME to provide reference for the Doc. no. & rev. for the load list considered here	Noted and vendor documents added to ref. list.	
5.	Appendix-1.2	1) Load not matching 2) Derive steps (numerous times same comments provided)	1)noted and updated as per verified and confirmed with COMPANY. 2)Noted and updated in appendix-1.1.	
6.	Appendix-1.2	Load cycle (load & Current, both) & further calculation will be updated based on the correct rating as commented	Noted and updated as per verified and confirmed ratings with COMPANY.	

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S. No.	Reference / Page No.	COMPANY Comments	CONTRACTOR Response	Status / Remarks
7.	Appendix-1.2	Also clarify the basis of 110V DC selected, is it defined in the UPS Datasheet??	Please note that battery voltage accepted refer to. "TQ-P30377-RSME-ELECTRICAL-0027".	
8.	Appendix-1.2	For same 110V DC Voltage, in few cal. RSME has used 95V (min. allowed DC Voltage) to evaluate the load current. In this Cal. nominal DC is used. Not acceptable	Noted and updated based on verified and confirmed ratings with COMPANY.	
9.	Appendix-1.3	Attach Vendor reference (as annexure) for the same	Noted and attached.	
10.	Appendix-1.3	It is battery, DC	Noted and updated.	
11.	Appendix-1.4	Conclude for the Boost Charge Compare Float & Boost and select worst	Noted and updated.	
12.	Appendix-2.1	Load list shall be based on the final vendor data RSME to provide reference for the Doc. no. & rev. for the load list considered here	Noted and vendor documents added to ref. list. However DWDM cabinet final load updated as per final load received from vendor after cabinet FAT refer to vendor mail attached to the CRS.	

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S. No.	Reference / Page No.	COMPANY Comments	CONTRACTOR Response	Status / Remarks
13.	Appendix-2.2	Derive Steps	Pplease note that load reflected directly form load list as the basis input for calculation.	
14.	Appendix-2.2	Refer all comments provided for the RDS-6 and update this calculation accordingly Rating will change and subject to approval	Noted and all comments incorporated and all ratings updated based on verified and confirmed ratings with COMPANY.	
15.	Appendix-3		Noted and vendor Kt & Tt factor tables added.	
ORIGINATOR		REVIEWED BY		APPROVED BY
Name		MD/SR		Senthivel Gopal
Sign				
Date		06-11-2025		06-11-2025

Mostafa Ahmed

Subject: FW: SE FEilds: Power calculations for PKG-2

From: Shashi Preetham Dontha <shashi@elcome.com>

Sent: Monday, 22 September 2025 12:14 PM

To: Rahul Prafulla Khandekar <r.prafulla@robtstone.ae>

Cc: Sandeep Zutshi <sandeep@elcome.com>; Akshit Kakkar <akshit@elcome.com>; P. K. Surendran <pkSURENDRAN@robtstone.ae>; Hemant Veen <hemant@robtstone.ae>

Subject: RE: SE FEilds: Power calculations for PKG-2

CAUTION: This E-Mail originated from outside of the organization. Please do not click links or open attachments unless you recognize the source of this email and knows that the content is safe.

Dear Mr. Rahul,

As discussed over the phone, please find below the revised power consumption table including RTU & KVM.

Kindly note the following:

- **Typical Power Consumption** reflects the power consumed under normal operating conditions at 25°C.
- In the event of an **HVAC failure**, if the cabinet temperature rises above 25°C, the fans inside the DWDM chassis will run at higher speeds, which increases the power consumption up to the **Maximum Power Consumption** values indicated in the table. To safeguard the equipment, it will automatically power off if the internal temperature exceeds 55°C.

	Typical Power Consumption(W)	Max. Power Consumption(W)
ASAB CDS 1	2861	5310
ASAB CDS 2	2821	5270
ASAB RDS1	1018	2137
ASAB RDS2	993	2111
ASAB RDS3	993	2111
ASAB RDS4	1018	2138
ASAB RDS5	993	2111
ASAB RDS6	1110	2210
ASAB GRS1	993	2111
ASAB GRS2	993	2111
ASAB GRS3	993	2111
ASAB GRS4	993	2111
ICC building	1037	2162
33kv SS	993	2111
Sahil CDS	1913	3751
Sahil RDS1	1071	2198

Thanks and regards,

This calculation shall be aligned in line with Vendor's approved (Code-1) UPS sizing calculation
Submit next Rev. as final upon Code-1 approved vendor sizing for both SAHIL area



**Abu Dhabi Company for Onshore Petroleum Operations Ltd.
(ADNOC ONSHORE)**

EPC FOR UPGRADATION OF FACILITIES AT SE FIELD

AGREEMENT NO.: 4700022908

COMPANY PROJECT NO.: P30377

TELECOM UPS SIZING CALCULATION - SAHIL AREA

P14429-16-99-67-1601

This document is Newly developed for this project

This document is prepared for the following projects and the project number P30377 is for common deliverables.

PROJECT DESCRIPTION	PROJECT NUMBER
EPC FOR UPGRADATION OF FACILITIES AT SE FIELD	P30377
REPLACEMENT OF AGED ELECTRICAL EQUIPMENT AT SE FIELD	P30255
REPLACEMENT OF SDH NETWORK AND TELECOM UPS AT ASAB/SAHIL	P14429
EPC SCOPE OF WORK FOR PHYSICAL SECURITY RECOMMENDATIONS PROJECT	P30297
REPLACEMENT OF WATER DISPOSAL GRE LINES AT SHAH	P29171
REPLACEMENT OF EXISTING ESP POWER SKIDS (24 NOS) AT SHAH	P29179
REQUIREMENT OF WHCP TO OPERATE DOWN HOLE SSSV AT SHAH	P29175

		M.DERAZ			
B	06-11-2025	MD/SR (SEE)	SG (LEE)	RPK (EM)	Issued for Approval
A	30-07-2025	MD/SR (SEE)	SG (LEE)	RPK (EM)	Issued for Review
REVISION	DATE	PREPARED	REVIEWED	APPROVED	DESCRIPTION

THIS DOCUMENT IS INTENDED FOR USE BY COMPANY AND ITS NOMINATED CONSULTANTS, CONTRACTORS, MANUFACTURERS AND SUPPLIERS

EPC CONTRACTOR



ADNOC ONSHORE DOCUMENT NO. 16-99-67-1601

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Class:2

ADNOC ONSHORE AGREEMENT NO.
4700022908

ADNOC ONSHORE
PROJECT NO. P30377

REV. B

DATE: 06-11-2025

TELECOM UPS SIZING CALCULATION - SAHIL AREA	Project No.: P14429		
	ADNOC Onshore Doc. No.: 16-99-67-1601		
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1 GENERAL

1.1 Introduction

ASAB, SAHIL & SHAH fields are the oil fields located at southeast of Abu Dhabi city and are operated by Abu Dhabi COMPANY for Onshore Petroleum Ltd (COMPANY). SAHIL, ASAB, & SHAH (SAS) fields are located in the desert area approximately 100 km, 160 km and 220 km, respectively from the Abu Dhabi city - see Figure 1.

Bu-Hasa field is located 200 kilometres southwest of Abu Dhabi as shown in figure 1. It is ADNOC Onshore's most western producing field. It was discovered in 1962 and was put on production in May 1965. In Bu-Hasa currently there are three Remote Degassing Stations (RDS-1/2/3) and two satellite stations BQ & HW.

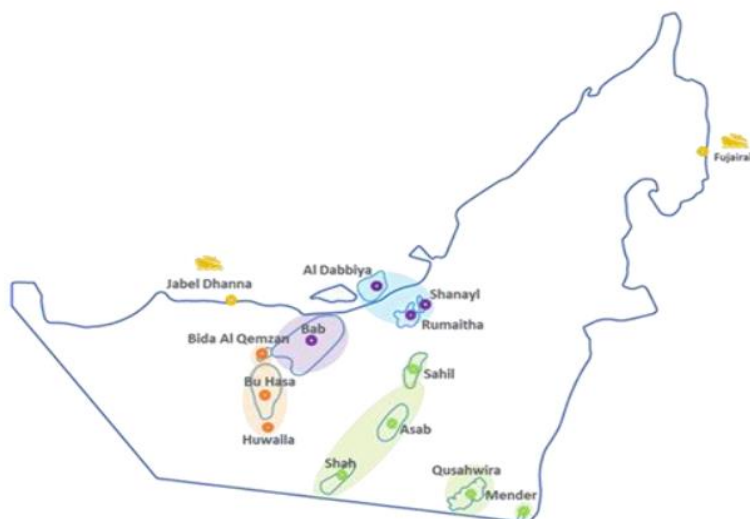


Figure-1: ADNOC ONSHORE FIELDS

The COMPANY awarded the Engineering, Procurement and Construction works for "Upgradation of Facilities at SE Fields" Project to M/s ROBT STONE (ME) LLC- as EPC-contractor.

1.2 Scope

The Scope of Work for EPC CONTRACTOR to carry out Modification Works at South East & Bu Hasa to implement for the purposes of maintaining continuous improvement for the existing facilities and concerns raised by operations, it is intended carrying modifications to improve the operability and safety of the existing facilities; consisting of modifications and upgradations of existing facilities, enhance the fault recovery of transmission system and enhance the security system.

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The Scope of work comprises the following packages:

- i. Package 1: Replacement of Aged Electrical Equipment at SE Fields.
 - a. ASAB
 - b. SAHIL
 - c. SHAH
 - d. Sub package 1: Replacement of existing ESP power skids(24Nos) at Shah
 - e. Sub package 2: Requirement of WHCP to operate downhole SSSV in Shah
- ii. Package 2: Replacement of SDH Network and Telecom UPS at Asab and Sahil.
- iii. Package 3: Physical Security Recommendation at Bu Hasa and SE Fields.
- iv. Package 4: Replacement of Water Disposal GRE Lines at SHAH.

1.3 Purpose of this Document

The purpose of this document is to define the technical basis, parameters and criteria for Telecom AC UPS Sizing Calculation in SAHIL area (SAHIL RDS-1 & SAHIL ADMIN BUILDING) for the “EPC FOR UPGRADATION OF FACILITIES AT SE FIELD Project.”

1.4 Definition and Glossary

CONTRACT NUMBER	: 4700022908
PROJECT NUMBER	: P30377
PROJECT	: EPC FOR UPGRADATION OF FACILITIES AT SE FIELD
COMPANY	: Abu Dhabi Company for Onshore Petroleum Operations Ltd (ADNOC Onshore)
EPC CONTRACTOR / CONTRACTOR	: Robt. Stone ME (RSME)
SUBCONTRACTOR	: Agency / Entity that will be engaged by Contractor in agreement with Company to perform specific work during EPC stage
MANUFACTURER / SUPPLIER / VENDOR	: The person, firm, corporation including their assignees to whom the purchase order is placed by CONTRACTOR.

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SHALL	:	The use of the word “shall” indicate a mandatory requirement
SHOULD	:	Indicates a strong recommendation to comply with the requirement the document
MAY	:	Indicates preference course of action

1.5 Abbreviation

Abbreviations		Description
ADNOC	:	Abu Dhabi National Oil Company (ADNOC Onshore)
AC	:	Alternating Current
DBM	:	Design Basis Memorandum
DETC	:	De-energised Tap Changer
ACDB		AC Distribution Board
DEP		Design and Engineering Practice
EMC	:	Electromagnetic Compatibility
ECMS	:	Electrical Control and Monitoring System
EN	:	European Norm
EPC	:	Engineering, Procurement and Construction
ES	:	Engineering Standard
FAT	:	Factory Acceptance Test
F&G	:	Fire & Gas
HV	:	High Voltage
HSE	:	Health, Safety and Environment
IP	:	Ingress Protection
IEC	:	International Electrotechnical Commission
IOC	:	International Oil Companies
ISO	:	International Organization for Standardisation
ITP	:	Inspection Test Plan
kV	:	Kilo Volts
kVA	:	Kilo Volt Amperes (Apparent Power)
kW	:	Kilo Watt
LED	:	Light Emitting Diode

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Abbreviations		Description
MCB	:	Miniature Circuit Breaker
MVA	:	Medium Voltage
OEM	:	Original Equipment Manufacturer
QA/QC	:	Quality Assurance/Quality Control
QMS	:	Quality Management System
SAT	:	Site Acceptance Test
SLD	:	Single Line Diagram
TBE	:	Technical Bid Evaluation.
TD	:	Technical Deviation
TQ	:	Technical Query
UPS	:	Uninterruptible Power Supply

1.6 Reference Code and Standards

The services shall be carried out in accordance with good engineering practices, latest revisions of all COMPANY specifications and recognized international codes of practices, standards and specifications along with their addenda applicable for the PROJECT.

1.6.1 Project Documents

Document Number	Title
P14429-30-99-93-0601 Rev.4	Scope of Work
P30377-30-99-91-1673	Electrical Design Basis
P30377-30-99-90-1603	Control Of Non-Conformities and Corrective Action Procedure (NCR)
P30377-30-99-90-1608	Procedure for Equipment / Material Criticality Rating
P30377-30-99-90-1604	Material Handling, Storage Control and preservation Procedure
P30377-30-99-97-1601	Project Quality Plan
P30377-30-99-90-1605	Vendor Inspection Procedure

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P30377-30-99-90-1613	Project Procurement Procedure
P30377-30-99-90-1612	Project Control Procedure
P14429-16-97-56-1601	SINGLE LINE DIAGRAM 25kVA, 240V AC TELECOM-UPS SYSTEM SAHIL ADMIN BUILDING
P14429-16-11-56-1601	SLD 40kVA, 240V/125V AC/DC TELECOM-UPS SYSTEM SAHIL RDS-1
P14429-30-99-53-1601	Specification for AC UPS
P14429-30-99-53-1602	Specification for AC DC UPS
P14429-30-99-54-1601	DATA SHEET FOR AC UPS
P14429-16-97-67-3701	UPS & BATTERY SIZING CALCULATION FOR 25kVA, 240V 1ph INDUSTRIAL AC UPS SAHIL ADMIN CDS 16-97 -UPS-4414
P14429-16-11-67-3701	UPS & BATTERY SIZING CALCULATION FOR 40kVA, 240V AC UPS SAHIL RDS-1 16-11-UPS-4402
P14429-30-99-74-3603	POWER CONSUMPTION AND HEAT DISSIPATION CALCULATION (PACKAGE 2)

1.6.2 AGES and ADNOC Onshore Standards

Document Number	Title
AGES-GL-13-001 Rev.1	ADNOC Group Projects & Engineering Contractors QA/QC Requirements
AGES-SP-13-001 Rev 1	Criticality Rating Specification
AGES-SP-13-002 Rev 1	ADNOC Group Projects & Engineering Procurement inspection and Certification Requirements in Projects
AGES-SP-07-004 Rev.1	Painting & Coating Specification

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1.6.3 Shell DEP's & Specification

Shell DEPs Version 43 to be followed :

Document Number	Title
DEP 33.64.10.10-Gen.	Electrical Engineering Design
DEP 33.65.40.31-Gen	Static Uninterruptible Power Supply (Static UPS Unit)
DEP 63.10.08.11-Gen.	Field commissioning and maintenance of electrical installations and equipment
DEP 33.64.10.17-Gen.	Application of Protective Functions for Electrical Systems.
DEP 33.64.10.33-Gen	Electromagnetic Compatibility (EMC) Requirements

1.6.4 International Codes and Standards

The International Codes, Standards and Recommended Practices which shall be utilised by CONTRACTOR for the execution of the works are referenced in the project documents.

Doument Number	Title
ISO 9001	Quality Managements System Requirements
ISO 9000	Quality Management System-Fundamentals and Vocabulary
ISO 19011	Guidelines for Auditing Management Systems
ISO 10006	QMS-Guidelines for Quality Management in Projects.
ISO 10005	QMS-Guidelines for Quality Plans
ISO/IEC 17020	General Criteria for the Operation of Various Types of Bodies Performing Inspection
ISO/IEC 17025	General Requirement for the Competence of Testing and Calibration Laboratories
ISO 27001	Information Security Management Systems
ISO/TR 10013	Guidelines for Quality Management System Documentation
IEC 60038	IEC Standard Voltages
IEC 60617-DB	Graphical symbols for diagrams.

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Doument Number	Title
IEC 60870	Tele-control Equipment and System.
IEC 61000	Electromagnetic Compatibility
IEC 61850	Communication networks and systems for power utility
IEC 61869	Instrument Transformers - All Parts.
IEC 62061	Safety of machinery - Functional safety of safety-related electrical,
IEC 60038	IEC Standard Voltages
IEEE 446	IEEE Recommended Practice for Emergency and standby power systems for Industrial and commercial applications
IEEE 946	IEEE Recommended Practice for the Design of Auxiliary Power systems for Generating stations
IEEE 1115	Recommended Practice for Sizing Nickel-Cadmium Batteries for Stationary Applications
IEEE 1184	Guide for Batteries for Uninterruptible Power Supply Systems

1.6.5 Order of Precedence

All design and engineering shall be performed in accordance with the latest Specifications, Standards, Codes, Regulations, and latest Shell DEPs.

In any areas of conflict, the order of precedence in the use of the revisions of the above as at the effective date of the Contract shall be as follows:

- a) The Laws, Standards and Regulations of the United Arab Emirates.
- b) ADNOC Code of Practice, HSE manuals and polices.
- c) Project Scope of Work.
- d) Project specific data sheets / specifications including ADNOC AGES Standards.
- e) COMPANY Procedures and Codes/Standards.
- f) COMPANY Specifications and Engineering Practices.
- g) COMPANY amendments and supplements to Shell DEPs.
- h) Shell DEPs (v43).

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- i) international oil and gas industry Codes, Standards and Recommended Practices (all where specified in the above or, where none of the above is applicable, as proposed by Engineering Contractor and approved by COMPANY).
- j) Internationally recognized sound practices in oil and gas industry.

In cases of conflict between documents in the same level of the hierarchy, the most stringent requirement shall apply. In such cases, CONTRACTOR shall provide its interpretation in writing of the most stringent requirement for COMPANY's approval utilizing a Technical Query sheet. In all cases of conflict, COMPANY's decision shall be final.

2 QUALITY ASSURANCE

EPC Contractor shall operate a Quality Management System in compliance with ISO 9001 Ed 2015, and AGES-GL-13-001 Rev 01 in all stages of Project.

Quality Control / Inspection & test plan shall be prepared as per Criticality rating & inspection class of the materials in compliance to Spec AGES-SP-13-001 Rev 01 & AGES-SP-13-002 Rev 01, for Procurement Inspection and Certification Requirement in Projects.

COMPANY & EPC CONTRACTOR, or Appointed Representative, reserves the right to carry out Quality Audits as defined in EPC Contractor Project Quality Plan to ensure that the Quality System requirements are being met.

3 SAFETY

CONTRACTOR shall prepare and maintain a project specific HSE plan that identifies and schedules all HSE activities of the project clearly identifying the HSE deliverables for all project phases in accordance with COMPANY Guidelines for the Preparation of HSE Plans This shall be a live document and shall be updated to include additional studies as well as schedule for all HSE activities during each stage by the CONTRACTOR.

4 ENVIRONMENT (UAE WEATHER)

4.1 Climatic Conditions

Wind		
i)	Prevailing Direction	From N-NW
ii)	Mean Speed	8 m/s
iii)	Basic Wind Speed (3 second gust)	44.7 m/s (as per ASCE-7 definition)
iv)	Basic Wind Speed (Hourly Mean Wind Speed)	26 m/s (as per BS 6399 definition)

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v)	Airborne Particles	Saliferous, sulfurous, dust and sand with frequent sandstorms
Temperature		
i)	Maximum Solar	85 °C
ii)	Maximum Shade	58 °C
iii)	Average Shade Summer	36 °C
iv)	Average Shade Winter	22 °C
v)	Average Shade Yearly	28 °C
vi)	Minimum Shade	4 °C
vii)	Maximum Soil at 1m depth	38 °C
viii)	Minimum Soil at 1m depth	13 °C
Humidity		
i)	Relative Maximum at 43 °C	95 %
ii)	Relative Maximum at 54 °C	60 %
Solar Radiation		946 W/m ² 1150-1250 W/m ² for OHL design
Rainfall		
i)	Frequency	Infrequent
ii)	Maximum	51 mm/year
iii)	Minimum	Trace
iv)	Highest rate	January-April, 25 mm in 24 hrs.
Dew		Heavy (daily)
Mists		Early morning mists causing Evaporative cooling loss.
Soil Thermal Resistivity		2.5 °C m/W

4.2 Equipment Design Temperature (Under Shed or Indoor)

Electrical Equipment Design Temperature	55 °C (40 °C indoor)
Motors, Generators	55 °C
Instrument Equipment	60 °C
Electrical buried Cables (at 1m depth)	40 °C
Electrical Cables in Air, Duct or Tray	55 °C
Air Cooler – non-process	55 °C
Battery Design Temperature	4 °C

4.3 A/C Design Conditions

Outdoor design temperature (Summer)	50 °C dry bulb/ 30 °C wet bulb
Outdoor daily range	15 °C
Outdoor design temperature (Winter)	6 °C dry bulb
Indoor Design Conditions	24 °C dry bulb/ 50% R. Humidity
Ambient Temperature for A/C selection	52 °C

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5 VOLTAGE LEVELS, FREQUENCY AND NEUTRAL SYSTEM

OPERATION VOLTAGE LEVEL	VOLTAGE VARIATION	PHASE	FREQUENCY	FREQUENCY VARIATION
415V	± 10 %	3 Ph+N	50Hz	± 5 %
240V	± 10 %	1 Ph+N	50Hz	± 5 %
110V DC	± 10 %	-	-	-

6 TELECOM UPS SIZING CALCULATION FOR SAHIL AREA:

- Battery sizing shall be carried out as per IEEE 1115 guidelines.
- Vented Nickel Cadmium recombination type batteries shall be used.
- Aging factor of minimum 1.25, future spare margin of 25% and design margin of 10% shall be considered as per the project specification
- The battery Ah shall be sized to provide required capacity at minimum ambient temperature of 4°C.
- The temperature de-rating shall be applied on batteries for environmental conditions defined in site & environmental data (Section 4).

6.1 Basis of UPS Sizing:

- The UPS will cater the 240V AC loads mainly for Field cameras & package loads as shown in load schedule (Appendix 1.1 & Appendix 2.1) and 110V DC loads as listed in (Appendix 1.1).
- UPS sizing is done based on the 240 V AC and 110V DC load schedule.
- Battery autonomy time for UPS is 8 hours for AC loads and 30mins for DC loads. Type of batteries shall be vented Ni-Cd. Battery.

6.2 LOAD Profile:

Autonomy time shall be as follows:

- 30mins for 110V DC UPS loads as shown in load schedule (Appendix 1.1)
- 8 hours for 240V AC UPS loads as shown in load schedule (Appendix 1.1 & Appendix 2.1)

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7 RESULTS AND RECOMMENDATION

Area	SAHIL RDS-1	SAHIL ADMIN BUILDING
Rectifier Rating (A)	500A	350A
No. of Cells	84	86
Battery Rating (Ah)	910Ah	725Ah
UPS Rating	40kVA	25kVA

Notes:

- 1) The Battery size Updated as per vendor calculation. The Kt factor updated as per vendor selected battery characteristics.

8 APPENDIXES

- Appendix-1: AC UPS Sizing Calculation SAHIL RDS-1, Service Building
- Appendix-2: AC UPS Sizing calculation SAHIL Admin Building
- Appendix-3: Kt Factor Calculation
- Appendix-4: Temperature Correction Factor (Tf)

TELECOM UPS SIZING CALCULATION - SAHIL AREA	Project No.: P14429		
	ADNOC Onshore Doc. No.: 16-99-67-1601		
	Rev.: B	Date: 06-11-2025	
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Appendix-1: AC UPS Sizing Calculation SAHIL RDS-1, Service Building

why 2 ESD & compressors ??

Review this load, 0.5Hr
back up time and 0.3
load factor...!!

APPENDIX-1.1 AC UPS SIZING TELECOM SAHIL RDS-1 LOAD LIST																															
Sr. No.	Item Description	Load Duration (Hr)	Qty. (No./set)	Number of Devices Energized	Power Supply (DC/AC) Volts	Power Consumption (Watts)	p.f.	Power Consumption (VA)	Load Factor	Effective Load (VA)	30-Mints Load (VA)	8-Hour Load (VA)	Remarks																		
1 DC LOADS																															
1.1	LV Switchgear	0.5	1	1	110	300	1	300	1	300	300																				
1.2	ESD	0.5	1	1	110	180	1	180	1	180	180																				
1.3	ESD	0.5	1	1	110	180	1	180	1	180	180																				
1.4	SUBSTATION EMERGENCY LIGHTING	0.5	1	1	110	1000	1	1000	0.3	300	300																				
1.5	AIR COMPRESSOR	0.5	1	1	110	600	1	600	1	600	600																				
1.6	AIR COMPRESSOR	0.5	1	1	110	600	1	600	1	600	600																				
2 AC LOADS																															
2.1	DWDM Cabinet	8	1	1	240	2198	0.8	2748	1	2747.50		2747.50																			
2.2	FDF Cabinet	8	1	1	240	400	0.8	500	1	500.00		500.00																			
2.3	WSC Cabinet	8	1	1	240	400	0.8	500	1	500.00		500.00																			
2.4	CCTV System Main Cabinet	8	1	1	240	400	0.8	500	1	500.00		500.00																			
2.5	Field Camera	8	1	1	240	96	0.8	120	1	120.00		120.00																			
2.6	Field Camera	8	1	1	240	96	0.8	120	1	120.00		120.00																			
2.7	Field Camera	8	1	1	240	96	0.8	120	1	120.00		120.00																			
2.8	Field Camera	8	1	1	240	96	0.8	120	1	120.00		120.00																			
2.9	Field Camera	8	1	1	240	96	0.8	120	1	120.00		120.00																			
2.10	PAGA System Cabinet	8	1	1	240	1600	0.8	2000	0.8	1600.00		1600.00																			
2.11	Telecom Bay-1	8	1	1	240	400	0.8	500	0.8	400.00		400.00																			
Total											2160.0	6847.5																			
Total Continuous UPS Power Consumption (Without Future margin) for 110V DC Loads						2160.0	VA		Battery Load Calculation <table><tr><td></td><td>DC Loads</td><td>AC Loads</td></tr><tr><td>Load</td><td>3.2</td><td>10.5</td></tr><tr><td>P.F</td><td>1</td><td>0.8</td></tr><tr><td>Input Power</td><td>3.2</td><td>8.4</td></tr><tr><td>0-30 min</td><td>11.61</td><td></td></tr><tr><td>30min - 8 hours</td><td>8.40</td><td></td></tr></table>						DC Loads	AC Loads	Load	3.2	10.5	P.F	1	0.8	Input Power	3.2	8.4	0-30 min	11.61		30min - 8 hours	8.40	
	DC Loads	AC Loads																													
Load	3.2	10.5																													
P.F	1	0.8																													
Input Power	3.2	8.4																													
0-30 min	11.61																														
30min - 8 hours	8.40																														
Total Continuous UPS Power Consumption (Without Future margin) for 110V DC Load considering 91% Efficiency of Converter						2373.6	VA																								
Total Continuous UPS Power Consumption (with 25% Future Spare & 10% Design Margin) 110V DC Loads						3.2	kVA																								
Total Continuous UPS Power Consumption (Without Future margin) for240V UPS Loads considering 88% Efficiency of Converter						7781.3	VA																								
Total Continuous UPS Power Consumption (with 25% Future Spare & 10% Design Margin) for 240V AC Loads						10.5	kVA																								
Total Continuous UPS Power Consumption						13.7	kVA																								
Selected UPS Size as per project SOW						40	kVA																								
Note 1 The VA values are being arrived at considering 0.8 power factor as confirmed by Instrumentation.																															

Update entire load list
based on the final
vendor approved data
and correct in Next
Rev.

**APPENDIX-1.2 CONSOLIDATED LOAD
TELECOM SAHIL RDS-1**

LOAD PROFILE

Sr. No.	Description	(0min - 30min)		(30min - 480min)		REMARKS
		Number of Devices Energized	LOAD (kW)	Number of Devices Energized	LOAD (kW)	
	Total AC & DC Loads	17	11.61	11	8.40	0.8PF Considered for Battery Sizing

DC LINK VOLTAGE SELECTED = 110 Volts

Total Loads (0 - 30 Min.) = 11.61 kW

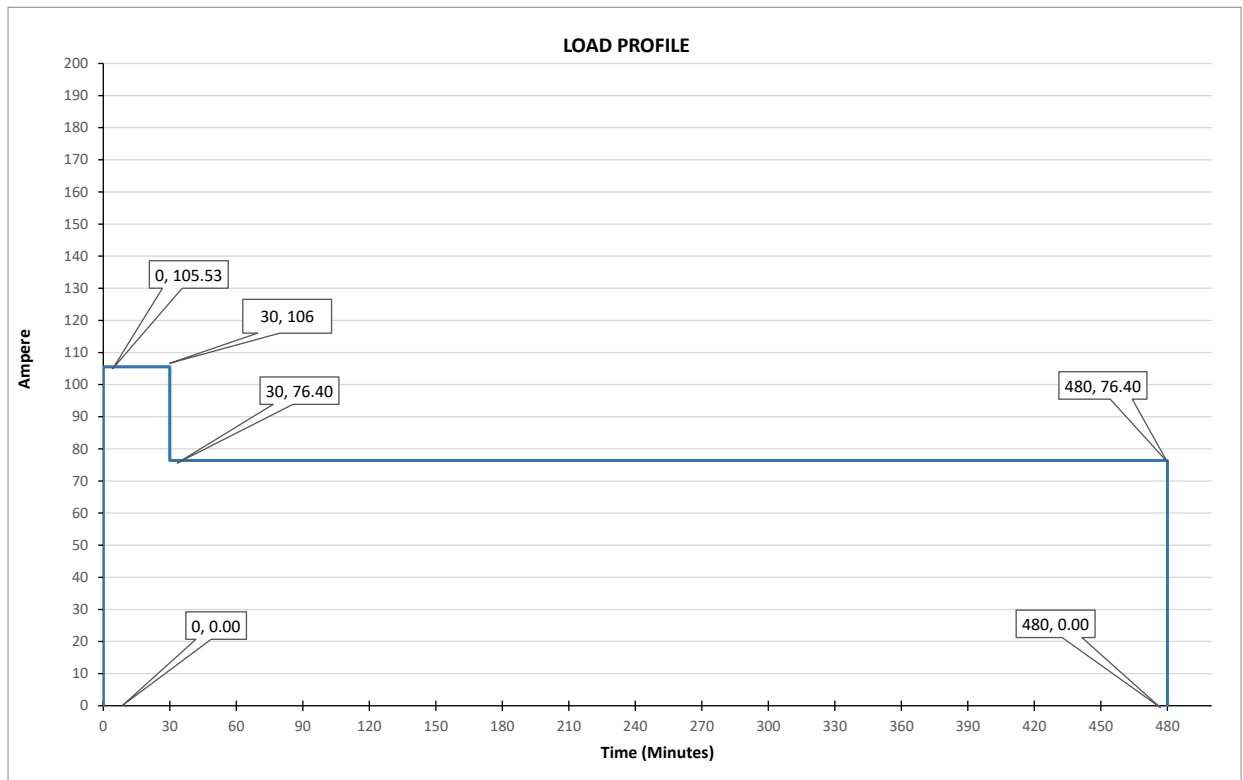
Total Current (0 - 30 Min.) = 105.53 A

Total Loads (30 - 480 Min.) = 8.40 kW

Total Current (30 - 480 Min.) = 76.40 A

For Battery Load cycle calculation, Loads are considered without margin.

Period	Time	Current	Duration
1	0 - 30 Min	105.53 A	30 Minutes
2	30-480 Min	76.40 A	450 Minutes



APPENDIX-1.3 BATTERY SIZING CALCULATION
TELECOM SAHIL RDS-1

CALCULATION OF NUMBER OF CELLS

DC link Voltage	=	110	Volts
Maximum Allowed Voltage	=	160	Volts (APPENDIX-1.5)
Minimum Allowed Voltage	=	92	Volts (APPENDIX-1.5)
End Cell Voltage Per Cell (End of Discharge Voltage)	=	1.095	Volts (APPENDIX-1.6)
Maximum Number of Cells	=	(Min. Allowed DC System Voltage) / (End Cell Voltage per Cell)	
	=	84.0	
Number of Cells Selected	=	84	
Minimum Available DC Voltage	=	(End of Discharge Voltage * Number of Cells)	
	=	92.0	Volts

update as per the final
selected vendor's
approved data

CALCULATION OF BATTERY AMPERE-HOUR

BASIS:

- 1 End discharge voltage is considered as 1.095 Volts.
- 2 Kt Factors have been taken from typical battery manufacturer catalogue.
- 3 Load amperes during the back-up time of 30Min & 8-Hour are picked from AC UPS load schedule
- 4 Tf Factors have been taken from typical battery manufacturer catalogue.
- 5 Battery Design Temperature 4°C

Section-1 First Period Only - If A2 > A1, go to Section-2 : No

Period	Specified Power (kW)	Load (Amp)		Change in Load (Amp)		Duration of Period (Minutes)		End of Section (Minutes)		Kt Factor	Temp. Derating Factor (Tf)	Required Section Size (Amp x Kt x Tf)
1	0.0	A1 =	105.53	A1 - 0 =	105.53	M1 =	30	t = M1 =	30	2.18	1.19	273.76

Total: 273.76

Section-2 First 2 Periods Only - If A3 > A2, go to Section-3 : No

Period	Specified Power (kW)	Load (Amp)		Change in Load (Amp)		Duration of Period (Minutes)		End of Section (Minutes)		Kt Factor	Temp. Derating Factor (Tf)	Required Section Size (Amp x Kt x Tf)
1	0.0	A1 =	105.53	A1 - 0 =	105.53	M1 =	30	t = M1+M2 =	480	8.1	1.1	940.26
2	0.0	A2 =	76.40	A2 - A1 =	-29.13	M2 =	450	t = M2 =	450	7.65	1.1	-245.14

Total: 695.12

The largest size section: = 695.12 Ampere-Hour

Considering Ageing Factor: = 1.25

Required battery size for the application: = 869 Ampere-Hour

Selected Battery capacity = 910 Ampere-Hour

APPENDIX-1.4 Rectifier and INVERTER SIZING **TELECOM SAHIL RDS-1**

INVERTER CURRENT (A):

$$L_c : \{(S \times PF \times 1000 \div V_{dc})\} \div \text{Inv. Eff.}_{\text{Batt. Float}}$$

where:

S	: UPS rating	=	40 kVA
Inv. Eff. _{Batt. Float}	: Inverter efficiency at battery float voltage	=	88 %
Float Voltage per Cell	: Battery Float Voltage per cell	=	1.42 V
V _{dc}	: number of cells x (Float voltage per cell)	=	119.28 V
L_c	: Inverter Current	=	304.8595 A

Rectifier Output Current (A):

$$I : [L_c + \{[(1.1 \times AH) \div T] \div \text{Inv. Eff.}_{\text{Batt. End}}\}] \times SF \quad (\text{As per IEEE 946})$$

where:

I	: minimum required charger rated output (A)		
1.1	: battery charger inefficiency factor	=	1.1
AH	: ampere-hour discharge from battery	=	910 AH
T	: time to recharge the battery (H)	=	10 H
Inv Eff Batt. End	: inverter efficiency at battery end voltage	=	88 %
SF	: Service Factor	=	1.15
LC	: Inverter Current	=	304.9 A

I	: Rectifier Output Current	=	481.40 A
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Boost Charging Mode

I _{boost}	: 0.1C5A (910 Ah x 0.1)	=	91.0 A
	: Continuous load current	=	105.5 A
I	: Rectifier Output Current (= Continuous load current + I _{boost})	=	196.5 A
Selected rectifier rating		=	500.0 A

IEEE 1115-2014 Ni-Cd Battery Sizing

Enquiry Details

Enquiry Number : ME17751-2
Date : March 03, 2025
Customer : SE
Project : EPC FOR UPGRADATION OF FACILITIES AT SE FIELD

Sizing Inputs (User Defined)

Nominal System Voltage : 110.00 V
Minimum System Voltage : 92.00 V
Maximum System Voltage : 160.00 V
Minimum Cell Voltage : 1.095 V
Number of Cells : 84 Nos
Minimum Temperature : 4.00 °C
Design Margin : 1.00
Aging Factor : 1.25
Cell Type : HVL

(*) Voltage Drop of 2.00 Volts is included in Minimum System Voltage

* Min. & Max.voltages are considered based on the Inverter input voltage tolerance.

As per Appendix-3 of Shell DEP 33.65.50.31-Gen, since autonomy time is greater than 20 mins, Vnom is considered for battery sizing.

Load Profile : Power ($I = P / V_{nom}$)

S. No.	Load (KW)	Duration (minutes)	Equivalent Amps
1	12.02	30.00	109.27
2	8.81	450.00	80.09

APPENDIX-1.6

Datasheet of HVL Battery

Enquiry Number : ME17751-2

Project Ref. : ADNOC - SAHIL RDS 1

Customer Name : SE

Revision : 0

S.No	Parameters	Battery Details
1	Battery type	VentPro Nickel Cadmium Pocket Plate Battery
2	Model offered	84 HVL 910
3	Battery Rating	910 Ah
4	No. of Cells	84 Cells, Cell Internal Resistance is 0.265 milli Ohms.
5	Cell dimensions in mm	304(L) x 195(W) x405(H) (1.2 V Block)
6	Recommended max. Period of storage	1 year in filled and charged condition
7	Capacity rated at	5 hour rate to 1.0 ECV at 20°C, End cell voltage is 1.095 V
8	Ah Efficiency	> 85%
9	Wh Efficiency	> 70%
10	Recommended float charging	Float charging current: 1 to 2 mA per Ah Float charging voltage: 1.42 V/Cell
	Recommended boost charging	Boost charging current: 0.1 C5 A Boost charging voltage: 1.45 V/Cell
11	Positive Plate type	Pocket plate
12	Negative Plate type	Pocket plate
13	Separator	Special Polypropylene Fibrous
14	Electrolyte	Potassium Hydroxide (KOH)
15	Container & Lid Material	Translucent Polypropylene
16	Container to lid sealing method	Heat Sealing Method
17	Safety vent	Valve Regulated vent with flame arrestor
18	Battery Rack/Stand (if required)	Mild Steel rack with Alkali Resistant Epoxy Powder Coating
19	Inter cell connectors	Nickel plated Copper
20	Nuts, Bolts and Washers for Inter cell connectors	Alkali Resistant Stainless Steel
21	Termination	Bolted type
22	Normal operating temperature range	-20°C to 40°C
23	Self discharge	2 to 3% per month at 20°C
24	Design float life	20 years at 20°C on float under recommended charging
25	Ventilation	Refer to ventilation report for air exchange
26	Battery shipment condition	Filled and discharged
27	Applicable standards	IEC 62259

TELECOM UPS SIZING CALCULATION - SAHIL AREA	Project No.: P14429		
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Appendix-2: AC UPS Sizing calculation SAHIL Admin Building

APPENDIX-2.1 AC UPS SIZING TELECOM SAHIL ADMIN BUILDING LOAD LIST																										
Sr. No.	Item Description	Load Duration (Hr)	Qty. (No./set)	Number of Devices Energized	Power Supply (AC) Volts	Power Consumption (Watts)	p.f.	Power Consumption (VA)	Load Factor	Effective Load (VA)	8-Hour Load (VA)	Remarks														
1	RF Equipment RACK	8	1	1	240	800	0.8	1000	1	1000.00	1000															
2	FO SYSTEM Rack	8	1	1	240	400	0.8	500	1	500.00	500															
3	MAIN DISTRIBUTION FRAME	8	1	1	240	400	0.8	500	1	500.00	500															
4	TELEPHONE ROOM DISTRIBUTION BOARD	8	1	1	240	80	0.8	100	1	100.00	100															
5	AMT FOR FIBER OPTIC SYSTEM	8	1	1	240	400	0.8	500	1	500.00	500															
1	TELEPHONE EXCHANGE	8	1	1	240	800	0.8	1000	1	1000.00	1000															
2	MAT FOR PAPX	8	1	1	240	400	0.8	500	1	500.00	500															
3	UHF BASE STATION	8	1	1	240	400	0.8	500	1	500.00	500															
4	WALL MOUNTED DISTRIBUTED FRAMW/ PATCH PANEL	8	1	1	240	400	0.8	500	1	500.00	500															
6	Field Camera	8	1	1	240	96	0.8	120	1	120.00	120															
7	Field Camera	8	1	1	240	96	0.8	120	1	120.00	120															
8	Field Camera	8	1	1	240	96	0.8	120	1	120.00	120															
9	Field Camera	8	1	1	240	96	0.8	120	1	120.00	120															
10	Field Camera	8	1	1	240	96	0.8	120	1	120.00	120															
	Total										5700.0															
Total Continuous UPS Power Consumption (Without Future margin) for 240V UPS Loads						5700.00	VA	<table><tr><th colspan="2">Battery Load Calculation</th></tr><tr><td></td><td>AC Loads</td></tr><tr><td>Load</td><td>7.70</td></tr><tr><td>P.F</td><td>0.8</td></tr><tr><td>efficiency</td><td>0.85</td></tr><tr><td>Input Power</td><td>7.24</td></tr><tr><td>0-480 min</td><td>7.24</td></tr></table>					Battery Load Calculation			AC Loads	Load	7.70	P.F	0.8	efficiency	0.85	Input Power	7.24	0-480 min	7.24
Battery Load Calculation																										
	AC Loads																									
Load	7.70																									
P.F	0.8																									
efficiency	0.85																									
Input Power	7.24																									
0-480 min	7.24																									
Total Continuous UPS Power Consumption (with 25% Future Spare & 10% Design Margin) for 240V AC Loads						7695.00	VA																			
Total Continuous UPS Power Consumption						7.70	kVA																			
Selected UPS Size						25.00	kVA																			
Note 1 The VA values are being arrived at considering 0.8 power factor as confirmed by Instrumentation.																										

**APPENDIX-2.2 CONSOLIDATED LOAD
TELECOM SAHIL ADMIN BUILDING**

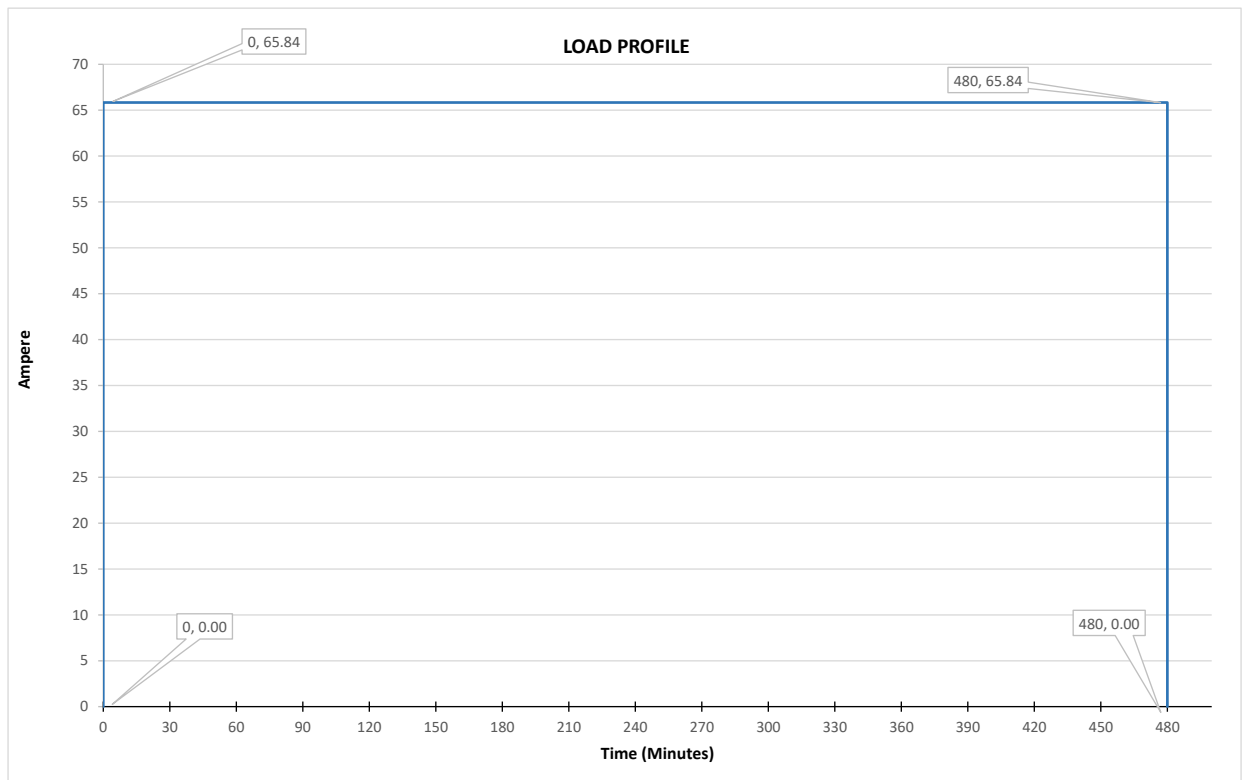
LOAD PROFILE

Sr. No.	Description	(0min - 480min)		REMARKS
		Number of Devices Energized	LOAD (W)	
	Total AC & DC Loads	14	7.24	0.8PF Considered for Battery Sizing

DC LINK VOLTAGE SELECTED = 110 Volts
 Total Loads (0 - 480 Min.) = 7.24 kW
 Total Current (0 - 480 Min.) = 65.84 A

For Battery Load cycle calculation, Loads are considered without margin.

Period	Time	Current	Duration
1	0-480 min	65.84 A	480 Minutes



APPENDIX-2.3 BATTERY SIZING CALCULATION
TELECOM SAHIL ADMIN BUILDING

CALCULATION OF NUMBER OF CELLS

DC link Voltage	=	110	Volts
Maximum Allowed Voltage	=	160	Volts (APPENDIX-2.5)
Minimum Allowed Voltage	=	92.11	Volts (APPENDIX-2.5)
End Cell Voltage Per Cell (End of Discharge Voltage)	=	1.071	Volts (APPENDIX-2.6)
Maximum Number of Cells	=	(Min. Allowed DC System Voltage) / (End Cell Voltage per Cell)	
	=	86.00	
Number of Cells Selected	=	86	
Minimum Available DC Voltage	=	(End of Discharge Voltage * Number of Cells)	
	=	92.11	Volts

CALCULATION OF BATTERY AMPERE-HOUR

BASIS:

- 1 End discharge voltage is considered as 1.071 Volts.
- 2 Kt Factors have been taken from typical battery manufacturer catalogue.
- 3 Load amperes during the back-up time of 8-Hour are picked from AC UPS load schedule
- 4 Tf Factors have been taken from typical battery manufacturer catalogue.
- 5 Battery Design Temperature 4°C

Section-1 First Period Only - If A2 > A1, go to Section-2 : No

Period	Specified Power (kVA)	Load (Amp)		Change in Load (Amp)		Duration of Period (Minutes)		End of Section (Minutes)		Kt Factor	Temp. Derating Factor (Tf)	Required Section Size (Amp x Kt x Tf)
1	0.0	A1 =	65.84	A1 - 0 =	65.84	M1 =	480	t = M1 =	480	7.98	1.1	577.94

Total: 577.94

The largest size section: = 577.94 Ampere-Hour

Considering Ageing Factor: = 1.25

Required battery size for the application: = 722.42 Ampere-Hour

Selected Battery capacity = 725 Ampere-Hour

APPENDIX-2.4 Rectifier and Inverter sizing TELECOM SAHIL ADMIN BUILDING

INVERTER CURRENT (A):

$$L_c : \{(S \times pf \times 1000 \div V_{dc})\} \div \text{Inv. Eff.}_{\text{Batt. Float}}$$

where:

S	: UPS rating	=	25 kVA
Inv. Eff. _{Batt. Float}	: Inverter efficiency at battery float voltage	=	91 %
Float Voltage per Cell	: Battery Float Voltage per cell	=	1.42 V
V _{dc}	: number of cells x (Float voltage per cell)	=	122.12 V
L _c	: Inverter Current	=	179.9707 A

Rectifier Output Current (A):

$$I : [L_c + \{[(1.1 \times AH) \div T] \div \text{Inv. Eff.}_{\text{Batt. End}}\}] \times SF \quad (\text{As per IEEE 946})$$

where:

I	: minimum required charger rated output (A)		
1.1	: battery charger inefficiency factor	=	1.1
AH	: ampere-hour discharge from battery	=	725 AH
T	: time to recharge the battery (H)	=	10 H
Inv Eff Batt. End	: inverter efficiency at battery end voltage	=	91 %
SF	: Service Factor	=	1.15
LC	: Inverter Current	=	180.0 A

$$I : \text{Rectifier Output Current} = 307.75 \text{ A}$$

Boost Charging Mode

I _{boost}	: 0.1C5A (725 Ah x 0.1)	=	72.5 A
	: Continuous load current	=	65.8 A
I	: Rectifier Output Current (= Continuous load current + I _{boost})	=	138.3 A
	Selected rectifier rating	=	350.0 A

Version : Beta 3.0

IEEE 1115-2014 Ni-Cd Battery Sizing

Enquiry Details

Enquiry Number : ME17751-1
Date : October 23, 2025
Customer : SE

Project : EPC FOR UPGRADATION OF FACILITIES AT SE FIELD

Sizing Inputs (IEEE 1115 Based)

Nominal System Voltage : 110.00 V
Minimum System Voltage *: 92.11 V
Maximum System Voltage : 160.00 V
Minimum Cell Voltage : 1.071 V
Number of Cells : 86 Nos
Minimum Temperature : 4.00 °C
Design Margin : 1.00
Aging Factor : 1.25
Cell Type : HVL

(*) Voltage Drop of 2.00 Volts is included in Minimum System Voltage

* Min. & Max.voltages are considered based on the Inverter input voltage tolerance.

As per Appendix-3 of Shell DEP 33.65.50.31-Gen, since autonomy time is greater than 20 mins, Vnom is considered for battery sizing.

Load Profile : Power ($I = P / V_{nom}$)

S. No.	Load (KW)	Duration (minutes)	Equivalent Amps
1	7.24	480.00	65.84

APPENDIX-2.6

Datasheet of HVL Battery

Enquiry Number : ME17751-1

Project Ref. :

Customer Name : SE

Revision : 0

S.No	Parameters	Battery Details
1	Battery type	VentPro Nickel Cadmium Pocket Plate Battery
2	Model offered	86 HVL 725
3	Battery Rating	725 Ah
4	No. of Cells	86 Cells Cell Internal Resistance is 0.332 milli Ohms.
5	Cell dimensions in mm	268(L) x 195(W) x405(H) (1.2 V Block)
6	Recommended max. Period of storage	1 year in filled and charged condition
7	Capacity rated at	5 hour rate to 1.0 ECV at 20°C End Cell Voltage is 1.071V.
8	Ah Efficiency	> 85%
9	Wh Efficiency	> 70%
10	Recommended float charging	Float charging current: 1 to 2 mA per Ah Float charging voltage: 1.42 V/Cell
	Recommended boost charging	Boost charging current: 0.1 C5 A Boost charging voltage: 1.45 V/Cell
11	Positive Plate type	Pocket plate
12	Negative Plate type	Pocket plate
13	Separator	Special Polypropylene Fibrous
14	Electrolyte	Potassium Hydroxide (KOH)
15	Container & Lid Material	Translucent Polypropylene
16	Container to lid sealing method	Heat Sealing Method
17	Safety vent	Valve Regulated vent with flame arrestor
18	Battery Rack/Stand (if required)	Mild Steel rack with Alkali Resistant Epoxy Powder Coating
19	Inter cell connectors	Nickel plated Copper
20	Nuts, Bolts and Washers for Inter cell connectors	Alkali Resistant Stainless Steel
21	Termination	Bolted type
22	Normal operating temperature range	-20°C to 40°C
23	Self discharge	2 to 3% per month at 20°C
24	Design float life	20 years at 20°C on float under recommended charging
25	Ventilation	Refer to ventilation report for air exchange
26	Battery shipment condition	Filled and discharged
27	Applicable standards	IEC 62259

TELECOM UPS SIZING CALCULATION - SAHIL AREA	Project No.: P14429		
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Appendix-3: K_t Factor Calculation

CIN : L40109TG1986PLC006745
D. No. 8-2-616, Plot No – 4,
Behind Minerva Grand Hotel,
Road No – 11, Banjara Hills,
Hyderabad – 500 034, T.S., India
Phone: +91-40-23305581;82 & 66167777
Fax : +91-40-23305580



Date: 31st October 2025

TO WHOM EVER IT MAY CONCERN

We hereby confirm that the standard Kt factor data are available for the end cell voltages (ECVs) of 1.14 V, 1.10 V, 1.05 V, and 1.00 V. In addition, based on the battery sizing calculations, we are also providing Kt factor data for the intermediate ECVs used in sizing the batteries.

HVL Battery Kt Factor Data:

ECV	MINUTES																			SECONDS		
	600	480	450	420	300	240	210	180	120	119	118	90	60	30	20	15	10	5	1	30	5	1
1.140	10.53	8.84	8.43	8.03	6.40	5.74	5.41	5.08	4.26	4.24	4.23	3.85	3.43	2.94	2.71	2.53	2.35	2.18	2.01	1.81	1.52	1.43
1.138	10.50	8.80	8.40	7.99	6.35	5.69	5.36	5.02	4.21	4.19	4.18	3.80	3.39	2.91	2.68	2.51	2.33	2.16	1.99	1.79	1.51	1.42
1.123	10.30	8.53	8.11	7.69	6.00	5.30	4.95	4.61	3.84	3.83	3.82	3.48	3.10	2.64	2.43	2.29	2.15	1.99	1.83	1.65	1.39	1.30
1.117	10.22	8.43	8.00	7.57	5.86	5.15	4.79	4.44	3.69	3.68	3.67	3.35	2.98	2.53	2.34	2.21	2.08	1.93	1.76	1.60	1.35	1.26
1.103	10.04	8.18	7.73	7.29	5.52	4.79	4.42	4.05	3.35	3.34	3.33	3.04	2.71	2.28	2.11	2.01	1.91	1.77	1.61	1.47	1.24	1.15
1.100	10.00	8.12	7.68	7.23	5.45	4.71	4.34	3.97	3.28	3.27	3.26	2.98	2.65	2.22	2.06	1.97	1.87	1.74	1.58	1.45	1.22	1.13
1.096	9.98	8.10	7.66	7.21	5.42	4.67	4.30	3.93	3.24	3.23	3.22	2.93	2.61	2.19	2.03	1.94	1.84	1.71	1.56	1.43	1.20	1.11
1.095	9.97	8.10	7.65	7.20	5.41	4.66	4.29	3.91	3.22	3.21	3.20	2.92	2.60	2.18	2.02	1.93	1.84	1.70	1.55	1.42	1.20	1.11
1.083	9.90	8.04	7.59	7.13	5.32	4.55	4.17	3.78	3.10	3.09	3.08	2.79	2.47	2.07	1.92	1.84	1.75	1.61	1.48	1.36	1.15	1.06
1.082	9.89	8.04	7.58	7.13	5.31	4.54	4.16	3.77	3.08	3.07	3.06	2.78	2.46	2.06	1.91	1.83	1.74	1.61	1.48	1.36	1.14	1.05
1.074	9.84	8.00	7.54	7.08	5.25	4.47	4.08	3.69	3.00	2.99	2.98	2.69	2.37	1.99	1.84	1.77	1.68	1.55	1.43	1.32	1.11	1.02
1.071	9.82	7.98	7.52	7.06	5.23	4.44	4.05	3.65	2.97	2.96	2.95	2.65	2.34	1.97	1.81	1.75	1.66	1.53	1.41	1.30	1.09	1.01
1.064	9.78	7.95	7.49	7.02	5.17	4.37	3.98	3.58	2.89	2.88	2.87	2.58	2.27	1.90	1.75	1.69	1.61	1.47	1.37	1.26	1.06	0.98
1.052	9.71	7.89	7.42	6.95	5.08	4.26	3.85	3.45	2.76	2.75	2.74	2.44	2.14	1.80	1.65	1.60	1.52	1.38	1.31	1.20	1.01	0.93
1.050	9.70	7.88	7.41	6.94	5.06	4.24	3.83	3.43	2.74	2.73	2.72	2.42	2.12	1.78	1.64	1.59	1.50	1.37	1.29	1.19	1.01	0.92
1.017	9.65	7.81	7.34	6.88	5.02	4.12	3.67	3.22	2.51	2.50	2.49	2.20	1.89	1.57	1.44	1.37	1.30	1.21	1.16	1.06	0.90	0.82
1.000	9.62	7.77	7.30	6.84	5.00	4.05	3.58	3.11	2.39	2.38	2.37	2.08	1.77	1.46	1.34	1.26	1.20	1.13	1.09	0.99	0.84	0.77

CIN : L40109TG1986PLC006745
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Phone: +91-40-23305581;82 & 66167777
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HVM Battery Kt Factor Data:

ECV	MINUTES																			SECONDS		
	600	480	450	420	300	240	210	180	120	119	118	90	60	30	20	15	10	5	1	30	5	1
1.140	10.18	8.24	7.75	7.26	5.32	4.34	3.84	3.35	2.57	2.56	2.55	2.23	1.92	1.54	1.39	1.32	1.21	1.05	0.93	0.86	0.75	0.71
1.133	10.14	8.20	7.71	7.22	5.28	4.30	3.81	3.32	2.51	2.50	2.49	2.16	1.84	1.48	1.34	1.26	1.16	1.02	0.89	0.83	0.72	0.68
1.124	10.07	8.14	7.66	7.17	5.23	4.26	3.77	3.28	2.44	2.43	2.42	2.07	1.74	1.40	1.27	1.20	1.10	0.97	0.85	0.79	0.69	0.65
1.100	9.91	8.00	7.52	7.03	5.10	4.14	3.65	3.17	2.25	2.23	2.22	1.83	1.47	1.20	1.09	1.02	0.94	0.84	0.74	0.69	0.60	0.57
1.050	9.81	7.86	7.38	6.91	5.02	4.06	3.57	3.09	2.11	2.10	2.08	1.63	1.25	0.95	0.85	0.80	0.73	0.66	0.61	0.57	0.50	0.47
1.000	9.69	7.83	7.36	6.89	5.00	4.03	3.55	3.06	2.09	2.07	2.05	1.59	1.11	0.80	0.71	0.66	0.60	0.54	0.52	0.48	0.43	0.40

The Kt factor for time t is calculated in the following table by interpolation using the formula:

$$Kt = Kt_2 - \frac{(t_2 - t)}{(t_2 - t_1)} (Kt_2 - Kt_1)$$

Nomenclature:

- K_t : interpolated value at t
- t: intermediate known value (e.g., time t where K_t is required)
- t_1, K_{t1} : first known point (e.g., time t_1 and K_{t1} factor value at t_1)
- t_2, K_{t2} : second known point (e.g., time t_2 and K_{t2} factor value at t_2)

Best Regards,



Maliyala Omkar
Senior Technical Manager
Head of Department - NCPP Application Engineering
HBL Engineering Limited (formerly known as HBL Power Systems Limited)

TELECOM UPS SIZING CALCULATION - SAHIL AREA	Project No.: P14429		
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Appendix-4: Temperature Correction Factor (T_f)

CIN : L40109TG1986PLC006745
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Fax : +91-40-23305580



Date: 4th November 2025

TO WHOM EVER IT MAY CONCERN

We hereby provide you the following temperature correction factors used in our Optimize Plus battery sizing calculations tool for VentPro Ni-Cd batteries.

HVL Battery Temperature Correction Factors:

Temp. in °C	Duration in Minutes									
	1 min	30 min	60 min	118	119	120 min	180	210	240	≥300 min
20	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
15	1.06	1.05	1.05	1.04	1.04	1.04	1.04	1.04	1.04	1.03
10	1.14	1.11	1.10	1.09	1.09	1.09	1.08	1.08	1.07	1.06
5	1.22	1.18	1.16	1.14	1.14	1.14	1.12	1.11	1.10	1.09
4	1.24	1.19	1.17	1.15	1.15	1.15	1.13	1.12	1.11	1.10
0	1.32	1.25	1.22	1.19	1.19	1.19	1.17	1.16	1.14	1.12
-5	1.46	1.38	1.33	1.29	1.29	1.29	1.25	1.23	1.21	1.18
-10	1.64	1.54	1.47	1.41	1.41	1.41	1.35	1.33	1.30	1.25

HVM Battery Temperature Correction Factors:

Temp. in °C	Duration in Minutes									
	1 min	30 min	60 min	118	119	120 min	180	210	240	≥300 min
20	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
15	1.07	1.05	1.04	1.03	1.03	1.03	1.03	1.03	1.02	1.02
10	1.14	1.11	1.08	1.07	1.07	1.07	1.06	1.06	1.05	1.04
5	1.23	1.17	1.13	1.11	1.11	1.11	1.09	1.08	1.08	1.06
4	1.25	1.18	1.14	1.12	1.12	1.12	1.10	1.09	1.08	1.07
0	1.33	1.24	1.18	1.15	1.15	1.15	1.13	1.12	1.11	1.09
-5	1.47	1.34	1.27	1.24	1.24	1.24	1.20	1.18	1.16	1.13
-10	1.64	1.47	1.39	1.33	1.33	1.33	1.27	1.25	1.23	1.18

Best Regards,

Maliyala Omkar
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